COMICCRAFT – AI COMIC STORY CREATOR USING GEMINI MODELS

5. Project Development Phase

5.1 Introduction

The Project Development Phase involves implementing the ComicCraft application based on the requirements and design prepared in the previous phases.

The frontend, backend, AI services, image generation, comic layout, and PDF export are developed and integrated during this phase.

5.2 Project Setup

The project is created using Python and FastAPI. The required folders and files are created according to the planned project structure.

The necessary Python packages are installed using the requirements file.

5.3 Backend Development

FastAPI is used to develop the backend application.

The backend manages user requests and communicates with the AI and image-generation services.

Important endpoints include:

- `/`
- `/health`
- `/generate`
- `/generate-comic/json`
- `/test-image`
- `/download`
- `/export-success`

5.4 Gemini Integration

Google Gemini models are integrated into ComicCraft to perform AI-based tasks.

The AI is used for:

- Understanding the user's story idea
- Creating the story outline
- Developing the story
- Generating narration
- Generating character dialogues
- Creating detailed image prompts

5.5 Image Generation

The image prompts generated by the AI are sent to the configured image-generation service.

The generated images are used as illustrations for the comic panels.

5.6 Frontend Development

The frontend provides a form where users can enter:

- Story Prompt
- Character Name
- Setting
- Tone
- Art Style

HTML and CSS are used for the interface, while JavaScript handles user interactions and communication with the backend.

5.7 Comic Layout

The generated story, dialogues, narration, and images are combined to create the final comic panels.

The panels are arranged in the required sequence for the final comic.

5.8 PDF Export

The completed comic is converted into a PDF using the PDF export functionality.

This allows the user to save and share the generated comic.

5.9 Conclusion

The Project Development Phase integrates all major components of ComicCraft into a working application. The final system can accept user input, generate AI-based comic content, create panels, display the comic, and export it as a PDF.